

MAT 310, Handout 19: Marginal and Conditional Distributions (5.3)

Marginal Probability Function of Discrete RVs: Given a joint probability function $p(y_1, y_2)$, the marginal probability functions of Y_1 and Y_2 are given by:

$$p_1(y_1) =$$

$$p_2(y_2) =$$

Example 1. The following table gives the joint probabilities of Y_1 and Y_2 .

		Y_2		
		0	1	2
Y_1	0	1/4	1/16	1/16
	2	1/8	0	1/8
	4	0	1/8	1/4

(a) Find $P(Y_2 = 1)$.

(b) Find $P(Y_1 = 0)$.

(c) Find the probability distribution of Y_1 .

(d) Find the probability distribution of Y_2 .

Conditional Probability Function (Discrete):

$$p(y_1|y_2) = P(Y_1 = y_1|Y_2 = y_2) = \frac{p(y_1, y_2)}{p_2(y_2)}$$

Note: defined only when $p_2(y_2) > 0$.

Example 2. Using the jointly distributed RVs in Example 1, find the following:

(a) $P(Y_2 = 1|Y_1 = 0) =$

(b) $P(Y_1 = 0|Y_2 = 1) =$

Let Y_1 and Y_2 be any RVs. The **conditional distribution function** of Y_1 given $Y_2 = y_2$ is

$$\begin{aligned} F(y_1|y_2) &= F(y_1|Y_2 = y_2) \\ &= P(Y_1 \leq y_1|Y_2 = y_2). \end{aligned}$$

Note: the conditional distribution function is defined for all RVs.

Marginal Density Function of Continuous RVs: Given a joint density function $f(y_1, y_2)$, the marginal probability functions of Y_1 and Y_2 are given by:

$$f_1(y_1) =$$

$$f_2(y_2) =$$

Conditional Density Function (Continuous): The conditional density of Y_1 given $Y_2 = y_2$ is given by

$$f(y_1|y_2) = \frac{f(y_1, y_2)}{f_2(y_2)}.$$

Note: defined only when $f_2(y_2) > 0$.

Conditional Distribution Function (Continuous case):

$$\begin{aligned} F(y_1|y_2) &= F(y_1|Y_2 = y_2) \\ &= P(Y_1 \leq y_1|Y_2 = y_2) \\ &= \int_{-\infty}^{y_1} f(t_1|y_2) dt_1 \\ &= \int_{-\infty}^{y_1} \frac{f(t_1, y_2)}{f_2(y_2)} dt_1 \end{aligned}$$

Similarly, $P(Y_1 \geq y_1|Y_2 = y_2) = \int_{y_1}^{\infty} \frac{f(t_1, y_2)}{f_2(y_2)} dt_1$.

Example 3. Suppose Y_1 and Y_2 have the following joint density function:

$$f(y_1, y_2) = \begin{cases} 3y_1, & 0 \leq y_2 \leq y_1 \leq 1 \\ 0, & o.w. \end{cases}.$$

(a) Find the marginal density function of Y_1 .

(b) Find the marginal density function of Y_2 .

(c) $P(Y_2 \geq 1/2) =$

(d) $P(Y_1 \geq 3/4 | Y_2 = 1/2) =$

(e) $P(Y_2 \leq 1/2 | Y_1 = 3/4) =$